Model Notes

Endogenous Education Choice in Segmented Search Markets

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Scope. This document contains the full mathematical description of the model: primitives, value functions, equilibrium conditions, solution algorithms, and transition dynamics. It serves as the companion reference for the paper and the codebase.

Changes relative to the previous draft. The following changes have been made relative to the March 11 draft:

1. **Parameter classification added** (Section 2). Parameters are now explicitly grouped into calibrated/fixed, pre-estimated, deep structural, and regime-specific, consistent with the paper plan. The demographic turnover rate ν is pre-estimated from CPS labour-force exit hazards rather than calibrated from a life-table assumption.
2. **Structure reorganised.** The document now follows a clearer progression: global environment $\rightarrow$ primitives $\rightarrow$ unskilled block $\rightarrow$ skilled block $\rightarrow$ global equilibrium $\rightarrow$ transition dynamics, with each block self-contained.
3. **Regime notation clarified.** Regime dependence is now written explicitly as (z) on all regime-specific objects. Deep parameters carry no regime subscript.
4. **Wage schedules collected.** All wage equations are now gathered in dedicated subsections within each block, making it easier to locate them.
5. **OJS wage premium derived.** The skilled OJS wage premium (eq. 28) is now stated explicitly.
6. **Transition dynamics unified.** Section 11 presents the full system of laws of motion for both segments jointly, including the aggregate mass dynamics and initial conditions, matching the paper plan's methodology section.
7. **Section on flow payoffs clarified.** b_U, b_T, b_S are now listed as deep (institutional) parameters, not regime-specific, consistent with the corrected classification.
8. **Matching efficiencies clarified.** μ_U, μ_S are now listed as deep (technology) parameters, not regime-specific.
9. **Unskilled vacancy HJB and free-entry corrected.** The vacancy HJB (11) and free-entry condition (12) now correctly normalise the expected firm value per contact by $1/\mathcal{U}_U(z)$, consistent with the fact that $u_U(x,z)$ is a mass density (not a probability density). The same correction is applied throughout the solution algorithm, model properties, and transition dynamics sections.
10. **Untrained employment accounting corrected.** The untrained employed density is now defined as $e_U(x,z) = m_U(x,z) - u_U(x,z) - t(x,z)$, where $m_U(x,z) = \ell(x) - m_S(x,z)$ is the total untrained mass. The previous definition $e_U = \ell - u_U - t$ did not account for the trained population m_S.
11. **Training cost functional form corrected.** The training cost is now $c(x) = (1-x)\,e^{(c-x)}$, matching the codebase. The parameter c enters through exponentiation.

Contents

1 Global environment

Time is continuous. All agents discount at rate $r > 0$.

1.1 Aggregate productivity and regimes

Aggregate productivity z_t is piecewise constant and takes values in a finite set $\mathcal{Z} = \{z_1, \ldots, z_K\}$. A *regime change* occurs at an (exogenous) stopping time T, at which z_t jumps from an initial value z_0 to a new value $z_1 \neq z_0$:

$$z_t = \begin{cases} z_0, & t < T, \\ z_1, & t \geq T. \end{cases}$$

The shock is *unexpected* (agents do not condition on an anticipated switch date) and *permanent* (once z switches, it remains at the new level forever).

1.2 Stationarity and transitions

For a fixed regime $z \in \mathcal{Z}$, the economy admits a stationary equilibrium (steady state) in which the cross-sectional distribution of workers across states is time-invariant. A regime change at time T generates a *transition path*: value functions and policies adjust immediately to the new regime z_1, while the cross-sectional distribution of workers is predetermined at T and evolves gradually according to flow-balance equations.

1.3 Sectoral productivity shifters

Each regime $z \in \mathcal{Z}$ maps into sector-specific productivity shifters $P_U(z) > 0$ and $P_S(z) > 0$. These shifters are constant within a regime and jump permanently at regime changes.

1.4 Demographic turnover

Workers exit the economy at Poisson rate $\nu > 0$ and are replaced one-for-one by newborn untrained workers with density $\ell(x)$. This delivers a stationary population composition within each fixed regime.

2 Parameter classification

Parameters fall into four groups, following the multi-regime estimation strategy described in the paper plan.

2.1 Calibrated / fixed

Symbol	**Name**	**Source / calibration**
r	Discount rate	4% annual real rate (risk-free rate)
ϕ	Training completion rate	NCES/IPEDS time-to-degree (1/48 months)

These are set externally and held fixed throughout all estimation stages.

2.2 Pre-estimated from data

Symbol	**Name**	**Source / method**
ν	Demographic turnover rate	CPS Basic Monthly LF→NILF exit hazard

The turnover rate ν is pre-estimated from matched CPS panels: $\hat{\nu}$ equals the average monthly labour-force exit rate for the working-age population (16–64) over the pre-FC baseline window. This avoids the collinearity with r that would arise if both were in the SMM objective, while grounding the value in observed transition data rather than a life-table assumption.

2.3 Deep structural parameters

Symbol	Name	Interpretation
a_ℓ, b_ℓ	Type distribution (Beta)	Worker heterogeneity
c	Training cost coefficient	Cost of education
η_U, η_S	Matching elasticities	CRS matching technology
μ_U, μ_S	Matching efficiencies	Matching technology level
β_U, β_S	Bargaining weights	Institutional wage setting
σ_S	OJS flow cost	On-the-job search cost
b_U	Unskilled UI flow value	Institutional UI policy
b_T	Training flow value	Training stipend/support
b_S	Skilled UI flow value	Institutional UI policy

These govern the *technology* and *institutions* of the labour market. Matching efficiencies (μ_U, μ_S) are properties of matching technology, invariant to the business cycle. Flow payoffs (b_U, b_T, b_S) reflect government policy institutions that do not necessarily change across regimes. All deep parameters are estimated once on the baseline window and held fixed thereafter.

2.4 Regime-specific parameters

Symbol	Name	Interpretation
$P_U(z), P_S(z)$	Productivity shifters	Aggregate state
$k_U(z), k_S(z)$	Vacancy costs	Hiring conditions
$\lambda_U(z), \lambda_S(z)$	Shock arrival rates	Match-destruction risk
$\alpha_U(z)$	Unskilled damage shape	Damage-shock distribution
$a_\Gamma(z), b_\Gamma(z)$	Skilled offer shape	Offer/quality distribution
$\xi_S(z)$	Exogenous skilled separation	Layoff rate

These capture the *aggregate state* of the economy and are re-estimated within each crisis window while holding deep parameters fixed.

3 Workers, firms, and types

3.1 Workers

Each worker has an innate type $x \in \mathcal{X} = [0, 1]$, fixed over time. Types are distributed according to CDF L with density ℓ ($x \sim \text{Beta}(a_\ell, b_\ell)$).

Market access via training (persistent eligibility). Workers can participate in either the *unskilled* market U (untrained status) or the *skilled* market S (trained status). Training changes *market access only*; it does not change innate type x.

Training decision at unskilled unemployment. Training can be initiated *only* by untrained workers who are in unskilled unemployment. For each regime z, the optimal policy is characterised by a cutoff $\bar{x}(z) \in [0, 1]$: upon entering unskilled unemployment in regime z, types $x > \bar{x}(z)$ initiate training, while types $x \leq \bar{x}(z)$ remain in unskilled search.

Interpretation under regime changes. Because training status is persistent, a regime change affects only the *flow into training* from the pool of untrained unemployed. After a regime change, the cutoff $\bar{x}(z)$ associated with the new regime governs the training decision of newly-arriving untrained unemployed workers, but the economy inherits the pre-shock stock of trained workers. Hence, after a switch (e.g. $z_L \to z_H$), the fraction trained adjusts gradually due to demographic turnover and equilibrium flows.

3.2 Firms

There are two labour markets. Both markets have homogeneous firms.

4 Training technology

Eligibility. Only untrained workers in unskilled unemployment may initiate training.

Cost. Initiating training requires a one-time cost $c(x)$ with $c'(x) < 0$. In applications,

$$c(x) = (1 - x)\, e^{(c-x)}.$$

Higher-ability workers face lower training costs. The parameter c enters through exponentiation so that the optimiser searches over $c \in \mathbb{R}$ while the cost coefficient $e^c > 0$ is guaranteed positive.

Completion. Training completes at Poisson rate $\phi > 0$. Upon completion, the worker becomes trained ($e = 1$) and enters skilled unemployment.

Flow payoffs. Let b_U, b_T, and b_S denote flow payoffs while unskilled unemployed, in training, and skilled unemployed, respectively. These are deep parameters (institutional), not regime-dependent.

Demographic turnover (stationarity device). Workers exit the economy at Poisson rate $\nu > 0$ and are replaced one-for-one by newborn untrained workers with density $\ell(x)$.

5 Production technologies

Unskilled production. In regime z, an unskilled match between worker type x and a homogeneous firm with match-specific efficiency $p \in (0, 1]$ produces

$$\pi_U(x, p, z) = P_U(z)\, x\, p.$$

Skilled production. In regime z, a skilled match between worker type x and a homogeneous firm with idiosyncratic quality $p \in (0, 1]$ produces

$$\pi_S(x, p, z) = P_S(z)\, x\, p.$$

6 Unskilled labour market

Fix a regime $z \in \mathcal{Z}$.

6.1 Matching

Let $\mathcal{U}_U(z)$ and $\mathcal{V}_U(z)$ denote the masses of unskilled unemployed and unskilled vacancies in regime z. Market tightness is

$$\theta_U(z) = \frac{\mathcal{V}_U(z)}{\mathcal{U}_U(z)}.$$

Matching follows a Cobb–Douglas function:

$$M_U(\mathcal{U}_U(z), \mathcal{V}_U(z)) = \mu_U\, \mathcal{U}_U(z)^{\eta_U}\, \mathcal{V}_U(z)^{1-\eta_U}, \qquad \eta_U \in (0,1),$$

implying vacancy filling and job-finding rates

$$q_U(\theta_U(z)) = \mu_U\, \theta_U(z)^{-\eta_U}, \qquad \theta_U(z)\, q_U(\theta_U(z)) = \mu_U\, \theta_U(z)^{1-\eta_U}.$$

6.2 Match productivity and shocks

New matches are created at frontier efficiency $p_0 = 1$.

Idiosyncratic damage shocks arrive at Poisson rate $\lambda_U(z) > 0$. Upon a shock, post-shock efficiency is

$$p' = e^{-d}, \qquad d \mid z \sim \mathrm{Exp}(\alpha_U(z)),$$

equivalently

$$p' \mid z \sim \mathrm{Beta}(\alpha_U(z), 1), \qquad G_z(p) = p^{\alpha_U(z)}, \quad p \in [0,1].$$

A match is endogenously destroyed whenever $p' < p^*(x,z)$ for a reservation threshold $p^*(x,z) \in (0,1)$.

6.3 Value functions and training policy

Let:

- $U_U(x,z)$ be the value of untrained unskilled unemployment,
- $U_U^{\text{search}}(x,z)$ the value of unskilled search,
- $T(x,z)$ the value of training,
- $E_U(x,p,z)$ the worker value of unskilled employment at efficiency p,
- $J_U(x,p,z)$ the firm value of a filled unskilled job,
- $V_U(z)$ the value of an unskilled vacancy.

Household values and training policy. Let $U_U(x,z)$ denote the value of *untrained* unskilled unemployment for type x in regime z. An untrained unemployed worker chooses between unskilled search and initiating training:

$$U_U(x,z) = \max\{U_U^{\text{search}}(x,z),\ U_U^{\text{train}}(x,z)\}, \qquad U_U^{\text{train}}(x,z) := -c(x) + T(x,z).$$

Training completes at Poisson rate ϕ and leads to skilled unemployment. The training value satisfies

$$(r + \phi + \nu)\, T(x,z) = b_T + \phi\, U_S(x,z). \tag{1}$$

Unskilled search yields flow payoff b_U and contacts vacancies at rate $\theta_U(z) q_U(\theta_U(z))$:

$$(r+\nu)\, U_U^{\text{search}}(x,z) = b_U + \theta_U(z) q_U(\theta_U(z))\big(E_U(x,1,z) - U_U^{\text{search}}(x,z)\big). \tag{2}$$

Define the net training gain

$$\Delta_T(x,z) := -c(x) + T(x,z) - U_U^{\text{search}}(x,z).$$

Under monotone incentives, there exists a cutoff $\bar{x}(z) \in [0,1]$ such that

$$\tau(x,z) = \mathbf{1}\{x > \bar{x}(z)\}, \qquad U_U^{\text{search}}(\bar{x}(z), z) = -c(\bar{x}(z)) + T(\bar{x}(z), z).$$

6.4 Match values, bargaining, and endogenous separation

A new unskilled match starts at frontier efficiency $p_0 = 1$. Idiosyncratic damage shocks arrive at rate $\lambda_U(z)$ and redraw efficiency p' from G_z on $[0, 1]$.

For $p \in (0, 1]$, worker and firm values satisfy

$$(r + \nu + \lambda_U(z))\, E_U(x, p, z) = w_U(x, p, z) + \lambda_U(z)\left[U_U(x, z)\, G_z(p^*(x, z)) + \int_{p^*(x,z)}^{1} E_U(x, p', z)\, \mathrm{d}G_z(p')\right], \tag{3}$$

$$(r + \nu + \lambda_U(z))\, J_U(x, p, z) = P_U(z)\, x\, p - w_U(x, p, z) + \lambda_U(z) \int_{p^*(x,z)}^{1} J_U(x, p', z)\, \mathrm{d}G_z(p'). \tag{4}$$

Wages are determined by Nash bargaining with worker weight β_U:

$$\beta_U\, J_U(x, p, z) = (1 - \beta_U)\big(E_U(x, p, z) - U_U(x, z)\big). \tag{5}$$

Endogenous separation is characterised by a reservation threshold $p^*(x, z) \in (0, 1)$ satisfying

$$J_U(x, p^*(x, z), z) = 0, \qquad J_U(x, p, z) \geq 0 \iff p \geq p^*(x, z). \tag{6}$$

6.5 Pointwise update for unskilled search values

Although the training option implies that the bargaining outside value is the full unemployment value $U_U(x, z) = \max\{U_U^{\text{search}}(x, z), U_U^{\text{train}}(x, z)\}$, the search value itself admits a convenient pointwise representation. Rearranging (2) gives

$$(r + \nu + \theta_U(z) q_U(\theta_U(z)))\, U_U^{\text{search}}(x, z) = b_U + \theta_U(z) q_U(\theta_U(z))\, E_U(x, 1, z). \tag{7}$$

Using Nash bargaining at entry efficiency $p_0 = 1$,

$$E_U(x, 1, z) = U_U(x, z) + \frac{\beta_U}{1 - \beta_U}\, J_U(x, 1, z),$$

we obtain the pointwise update

$$U_U^{\text{search}}(x, z) = \frac{b_U + \theta_U(z) q_U(\theta_U(z)) \left(U_U(x, z) + \frac{\beta_U}{1-\beta_U}\, J_U(x, 1, z)\right)}{r + \nu + \theta_U(z) q_U(\theta_U(z))}. \tag{8}$$

Under the cutoff policy $\tau(x, z) = \mathbf{1}\{x > \bar{x}(z)\}$,

$$U_U(x, z) = \begin{cases} U_U^{\text{search}}(x, z), & x \leq \bar{x}(z), \\ U_U^{\text{train}}(x, z), & x > \bar{x}(z), \end{cases}$$

so (8) becomes a simple piecewise update. In particular, on the search region $x \leq \bar{x}(z)$ it reduces to

$$(r + \nu)\, U_U^{\text{search}}(x, z) = b_U + \frac{\beta_U}{1 - \beta_U}\, \theta_U(z) q_U(\theta_U(z))\, J_U(x, 1, z).$$

6.6 Unskilled match surplus (wage-free equation)

Define the total match surplus as worker net gain plus firm value:

$$S_U(x, p, z) \;\equiv\; \big(E_U(x, p, z) - U_U(x, z)\big) + J_U(x, p, z).$$

Equivalently, $E_U(x, p, z) + J_U(x, p, z) = S_U(x, p, z) + U_U(x, z)$.

Adding the worker and firm HJBs (3)–(4) and substituting $E_U + J_U = S_U + U_U$:

$$(r+\nu+\lambda_U(z))\, S_U(x, p, z) = P_U(z)\, x\, p - (r+\nu)\, U_U(x, z) + \lambda_U(z) \int_{p^*(x,z)}^{1} S_U(x, p', z)\, \mathrm{d}G_z(p'). \tag{9}$$

6.7 Wage schedule (unskilled)

Given the surplus split $J_U = (1-\beta_U)S_U$ and $E_U - U_U = \beta_U S_U$, wages are recovered from the firm HJB (4). Multiplying the surplus equation (9) by $(1-\beta_U)$ and comparing with the firm HJB gives

$$w_U(x,p,z) = \beta_U\, P_U(z)\, x\, p + (1-\beta_U)(r+\nu)\, U_U(x,z). \tag{10}$$

The wage is linear in match quality p: the worker captures share β_U of flow output $P_U(z)\,x\,p$, plus a flow option value $(1-\beta_U)(r+\nu)U_U(x,z)$ reflecting the firm's outside value when the worker exits (at rate ν) or the match is destroyed and the worker returns to U_U.

6.8 Vacancies, free entry, and market tightness

Let $V_U(z)$ denote the value of an unskilled vacancy in regime z. Posting a vacancy costs $k_U(z)$ per unit time and yields contacts at rate $q_U(\theta_U(z))$.

Let $u_U(x,z)$ denote the density of untrained unskilled unemployed by type and define

$$\mathcal{U}_U(z) = \int_0^1 u_U(x,z)\,\mathrm{d}x.$$

A vacancy meets type x with probability $u_U(x,z)/\mathcal{U}_U(z)$ and creates a match at $p_0 = 1$, yielding value $J_U(x,1,z)$. The vacancy HJB is

$$rV_U(z) = -k_U(z) + q_U(\theta_U(z))\left(\frac{1}{\mathcal{U}_U(z)}\int_0^1 J_U(x,1,z)\,u_U(x,z)\,\mathrm{d}x - V_U(z)\right). \tag{11}$$

Free entry implies

$$V_U(z) = 0 \quad\Longrightarrow\quad q_U(\theta_U(z)) = \frac{k_U(z)\,\mathcal{U}_U(z)}{\int_0^1 J_U(x,1,z)\,u_U(x,z)\,\mathrm{d}x}, \tag{12}$$

which pins down market tightness $\theta_U(z)$ given the endogenous composition of unemployed workers.

6.9 Reservation rule

Setting $S_U(x,p^*,z) = 0$ in the surplus equation (9):

$$p^*(x,z) = \frac{(r+\nu)\,U_U(x,z) - \lambda_U(z)\,\int_{p^*}^1 S_U(x,p',z)\,\mathrm{d}G_z(p')}{P_U(z)\,x}. \tag{13}$$

6.10 Stationary cross-sectional distributions (unskilled)

Within a fixed regime z, the cross-sectional distribution of *untrained* workers across states is pinned down by flow balance. Let $u_U(x,z)$ denote the density of untrained workers in unskilled unemployment, $t(x,z)$ the density of workers in training, and define the untrained-segment mass by type:

$$m_U(x,z) \;\equiv\; \ell(x) - m_S(x,z),$$

where $m_S(x,z) = \frac{\phi}{\nu}\,t(x,z)$ is the trained mass (in stationarity). The untrained employed density is the residual:

$$e_U(x,z) \;:=\; m_U(x,z) - u_U(x,z) - t(x,z).$$

Training. At equilibrium the training distribution is stationary:

$$0 = \tau(x,z)\,u_U(x,z) - (\phi+\nu)\,t(x,z), \quad\Longrightarrow\quad t(x,z) = \frac{\tau(x,z)}{\phi+\nu}\,u_U(x,z). \tag{14}$$

Untrained unemployment. Untrained unemployment receives inflow from demographic replacement and job destruction, and loses mass due to job finding, training entry, and demographic exit. The density flow balance is

$$\Big(\theta_U(z)q_U(\theta_U(z)) + \tau(x,z) + \nu\Big)\, u_U(x,z) = \nu\, \ell(x) + \lambda_U(z)\, G_z(p^*(x,z))\, e_U(x,z), \tag{15}$$

where $e_U(x,z) = m_U(x,z) - u_U(x,z) - t(x,z)$.

7 Unskilled-block equilibrium and solution algorithm

Fix a regime $z \in \mathcal{Z}$. An *unskilled-block equilibrium* in regime z is a collection

$$\{\theta_U(z),\ U_U^{\text{search}}(\cdot,z),\ T(\cdot,z),\ U_U(\cdot,z),\ E_U(\cdot,\cdot,z),\ J_U(\cdot,\cdot,z),\ p^*(\cdot,z),\ u_U(\cdot,z),\ t(\cdot,z)\}$$

such that:

(E1) **Household values and training.** $T(x,z)$ satisfies (1); $U_U^{\text{search}}(x,z)$ satisfies (2) (equivalently (8)); and unemployment values satisfy $U_U(x,z) = \max\{U_U^{\text{search}}(x,z),\ -c(x) + T(x,z)\}$, $\tau(x,z) = \mathbf{1}\{-c(x) + T(x,z) \geq U_U^{\text{search}}(x,z)\}$.

(E2) **Match values and wages.** Worker and firm match values satisfy (3)–(4); wages satisfy Nash bargaining (5). (Equivalently, $w_U(x,p,z)$ is recovered from (10).)

(E3) **Endogenous separation.** The reservation threshold $p^*(x,z)$ satisfies (6).

(E4) **Vacancy creation (free entry).** Market tightness $\theta_U(z)$ satisfies free entry (12).

(E5) **Stationary cross-sectional distributions.** Densities satisfy the flow-balance conditions (14) and (15). Employment of untrained workers is the residual mass $e_U(x,z) = m_U(x,z) - u_U(x,z) - t(x,z)$, where $m_U(x,z) = \ell(x) - m_S(x,z)$.

7.1 Solution algorithm (unskilled side; fixed regime z)

For computation, it is convenient to organise the unskilled block as a two-layer fixed point. The *inner loop* solves the value functions $(U_U^{\text{search}}, U_U, T, E_U, J_U)$ given conjectures for market tightness θ_U and the reservation rule $p^*(\cdot)$. The *outer loop* updates the policy objects $(\tau(\cdot), p^*(\cdot), \theta_U)$ and the stationary composition (u_U, t) until all equilibrium conditions hold jointly.

Outer-loop unknowns. We iterate on

$$\Big(\theta_U^{(n)}(z),\ p^{*,(n)}(\cdot,z),\ \tau^{(n)}(\cdot,z),\ u_U^{(n)}(\cdot,z),\ t^{(n)}(\cdot,z)\Big).$$

Step 1. Initialise (outer loop). Choose initial guesses $\theta_U^{(0)}(z) > 0$ and a feasible reservation rule $p^{*,(0)}(x,z) \in (0,1)$ (e.g. constant in x). Initialise a training policy $\tau^{(0)}(x,z)$ (or cutoff $\bar{x}^{(0)}(z)$) and stationary densities $u_U^{(0)}(x,z)$ and $t^{(0)}(x,z)$ (e.g. proportional to ℓ).

Step 2. Inner value-function iteration (given $\theta_U^{(n)}$ and $p^{*,(n)}$). Holding $(\theta_U^{(n)}(z), p^{*,(n)}(\cdot,z))$ fixed, iterate jointly on $(U_U^{\text{search}}, U_U, T, E_U, J_U)$ until convergence. One convenient inner cycle is:

- **Step 2.1. Training value.** Update $T^{(m+1)}(x,z)$ from (1), treating $U_S(\cdot,z)$ as given by the skilled block.
- **Step 2.2. Unemployment value and training decision.** Given $T^{(m+1)}$ and the current search value $U_U^{\text{search},(m)}$, update the full outside value

$$U_U^{(m+1)}(x,z) = \max\{U_U^{\text{search},(m)}(x,z),\ -c(x) + T^{(m+1)}(x,z)\}$$

and the implied training policy

$$\tau^{(m+1)}(x,z) = \mathbf{1}\{-c(x) + T^{(m+1)}(x,z) \geq U_U^{\text{search},(m)}(x,z)\}$$

.

- **Step 2.3. Match values.** Given $U_U^{(m+1)}(x,z)$ and the fixed reservation rule $p^{*,(n)}(\cdot,z)$, solve the coupled system (3)–(4) together with bargaining (5) to obtain $E_U^{(m+1)}(x,p,z)$ and $J_U^{(m+1)}(x,p,z)$.
- **Step 2.4. Search value.** Update $U_U^{\text{search},(m+1)}(x,z)$ from the HJB (2):

$$(r+\nu)\, U_U^{\text{search},(m+1)}(x,z) = b_U + \theta_U^{(n)}(z) q_U(\theta_U^{(n)}(z))(E_U^{(m+1)}(x,1,z) - U_U^{\text{search},(m+1)}(x,z))$$

.

Stop the inner loop when the sup norm of updates to $(U_U^{\text{search}}, U_U, E_U, J_U)$ is below tolerance.

Step 3. Update stationary distributions (given the converged policy and hazards). Using the converged training policy $\tau^{(n+1)}(\cdot,z)$ from the inner loop and the current reservation rule $p^{*,(n)}(\cdot,z)$, update the stationary densities $(u_U^{(n+1)}, t^{(n+1)})$ by solving (14)–(15) pointwise in x (with $e_U(x,z) = m_U(x,z) - u_U(x,z) - t(x,z)$).

Step 4. Update reservation rule p^* (outer update). Setting $S_U(x,p^*,z) = 0$ in the closed-form surplus expression yields the direct formula (13). Anderson acceleration (rank-1) is applied to the vector $\{p^{*,(n+1)}(x,z)\}_x$ before writing it back to cache.

Step 5. Update tightness θ_U by free entry (outer update). Given the updated composition $u_U^{(n+1)}(\cdot,z)$ and the converged frontier surplus $J_U(x,1,z)$, update θ_U using free entry (12):

$$q_U(\theta_U^{(n+1)}(z)) = \frac{k_U(z)\,\mathcal{U}_U^{(n+1)}(z)}{\int_0^1 J_U(x,1,z)\, u_U^{(n+1)}(x,z)\,\mathrm{d}x},$$

and invert $q_U(\theta) = \mu_U\, \theta^{-\eta_U}$ to obtain $\theta_U^{(n+1)}(z)$. Use Anderson acceleration when updating tightness, to reduce fluctuations.

Step 6. Convergence (outer loop). Stop if

$$\max\Big\{|\theta_U^{(n+1)}(z) - \theta_U^{(n)}(z)|,\ \|p^{*,(n+1)}(\cdot,z) - p^{*,(n)}(\cdot,z)\|_\infty,\ \|u_U^{(n+1)}(\cdot,z) - u_U^{(n)}(\cdot,z)\|_\infty\Big\} < \text{tol}.$$

Otherwise set $n \leftarrow n+1$ and return to Step 2.

Remark 1 (Two-block model)**.** *In the full model, Step 2 takes $U_S(\cdot,z)$ as given from the skilled block, while the skilled block takes the trained mass $m_S(\cdot,z) = \frac{\phi}{\nu} t(\cdot,z)$ (in steady state) as given from the unskilled block. A global equilibrium is obtained by iterating on the two blocks until these link variables are consistent.*

8 Skilled labour market

This section adapts the Pissarides (2000) Chapter 4 framework ("Labour Turnover and On-the-Job Search") to our segmented environment with aggregate regime $z \in \mathcal{Z}$ and worker heterogeneity $x \in [0,1]$. The skilled block features (i) idiosyncratic match quality $p \in [0,1]$ that is redrawn by shocks, (ii) endogenous separation via a cutoff $p_S^*(x,z)$, and (iii) on-the-job search (OJS) as a discrete choice with a flow search cost.

8.1 Primitives and market objects

Trained population. Only trained workers participate in the skilled market. Let $m_S(x,z)$ be the density of trained workers of type x in regime z. In stationarity,

$$m_S(x,z) = \frac{\phi}{\nu}\, t(x,z),$$

as implied by training completion and demographic turnover.

Production. A skilled match between worker type x and idiosyncratic quality $p \in [0,1]$ produces $\pi_S(x,p,z) = P_S(z)\, x\, p$.

Separations and quality shocks. A filled skilled match is destroyed exogenously at Poisson rate $\xi_S(z) > 0$. In addition, match quality is hit by shocks at Poisson rate $\lambda_S(z) > 0$. Upon a shock, quality is redrawn i.i.d. from Γ_z on $[0,1]$ with density γ_z. A match is endogenously destroyed whenever the post-shock draw is below a reservation cutoff $p_S^*(x,z) \in (0,1)$.

Interpretation of Γ_z. The distribution Γ_z is the *match-quality distribution* in the skilled market. It governs both (i) initial match formation (offers to unemployed workers) and (ii) quality redraws after idiosyncratic shocks. Unlike the unskilled market, skilled matches are not born at a fixed frontier; instead, match quality is drawn from Γ_z at formation and after shocks. We parameterise $\Gamma_z = \mathrm{Beta}(a_\Gamma(z), b_\Gamma(z))$.

OJS (discrete choice). An employed skilled worker chooses whether to search while employed. Searching incurs a flow cost $\sigma_S \geq 0$ (paid by the worker). If searching, the worker receives outside offers at the same contact rate as an unemployed job seeker, $\theta_S(z)\, q_S(\theta_S(z))$. Each offer is a draw $\tilde{p} \sim \Gamma_z$. The worker quits to the outside job iff the offer is strictly better (job ladder), $\tilde{p} > p$.

Demographic turnover. All workers exit at Poisson rate $\nu > 0$.

8.2 Matching and tightness

Let $u_S(x,z)$ denote the density of skilled unemployed by type, and let $e_S(x,p,z)$ denote the density of employed trained workers by (x,p). Define the OJS policy $s^*(x,p,z) \in \{0,1\}$ (defined below), and the aggregate mass of skilled job seekers in regime z as

$$\widetilde{\mathcal{U}}_S(z) \;\equiv\; \underbrace{\int_0^1 u_S(x,z)\,\mathrm{d}x}_{\mathcal{U}_S(z)} + \underbrace{\int_0^1\int_0^1 s^*(x,p,z)\, e_S(x,p,z)\,\mathrm{d}p\,\mathrm{d}x}_{\mathcal{E}_S^{\text{search}}(z)}.$$

Let $\mathcal{V}_S(z)$ be the mass of skilled vacancies and define tightness as

$$\theta_S(z) = \frac{\mathcal{V}_S(z)}{\widetilde{\mathcal{U}}_S(z)}.$$

Matching is Cobb–Douglas in vacancies and job seekers:

$$M_S(\widetilde{\mathcal{U}}_S(z), \mathcal{V}_S(z)) = \mu_S\, \widetilde{\mathcal{U}}_S(z)^{\eta_S}\, \mathcal{V}_S(z)^{1-\eta_S}, \qquad \eta_S \in (0,1),$$

implying the vacancy contact (filling) rate

$$q_S(\theta_S(z)) = \mu_S\, \theta_S(z)^{-\eta_S},$$

and the contact rate for any job seeker (unemployed or employed searcher) $\theta_S(z)\, q_S(\theta_S(z))$.

8.3 Value functions, bargaining, and two wage schedules

Let:

- $U_S(x,z)$ be the value of skilled unemployment;
- $E^0_S(x,p,z)$ and $E^1_S(x,p,z)$ be worker values in a job of quality p without and with OJS;
- $J^0_S(x,p,z)$ and $J^1_S(x,p,z)$ be the corresponding firm values;
- $V_S(z)$ be the value of a skilled vacancy.

Define the worker's continuation value in a job of quality p as

$$E_S(x,p,z) \equiv \max\{E^0_S(x,p,z),\ E^1_S(x,p,z)\}, \qquad J_S(x,p,z) \equiv J_S^{s^*(x,p,z)}(x,p,z).$$

Surplus and Nash bargaining. For each search regime $s \in \{0,1\}$ define match surplus

$$S^s_S(x,p,z) \equiv \big(E^s_S(x,p,z) - U_S(x,z)\big) + J^s_S(x,p,z).$$

Nash bargaining with worker weight $\beta_S \in (0,1)$ implies

$$E^s_S(x,p,z) - U_S(x,z) = \beta_S\, S^s_S(x,p,z), \qquad J^s_S(x,p,z) = (1-\beta_S)\, S^s_S(x,p,z), \qquad s \in \{0,1\}. \tag{16}$$

This delivers two wage schedules $w^0_S(x,p,z)$ and $w^1_S(x,p,z)$ because the continuation values differ across s (search changes quit hazards and introduces the search cost).

Unemployed worker. A skilled unemployed worker meets a vacancy at rate $\theta_S(z)q_S(\theta_S(z))$ and draws $\tilde p \sim \Gamma_z$. The worker accepts iff the resulting match has nonnegative surplus (equivalently $\tilde p \ge p^*_S(x,z)$). New matches start with the worker free to choose whether to search or not, i.e. the acceptance value is $E_S(x,\tilde p,z)$. The unemployment HJB is

$$(r+\nu)\,U_S(x,z) = b_S + \theta_S(z)q_S(\theta_S(z)) \int_{p^*_S(x,z)}^{1} \big(E_S(x,\tilde p,z) - U_S(x,z)\big)\,\mathrm{d}\Gamma_z(\tilde p). \tag{17}$$

Write $\kappa_S(z) \equiv \theta_S(z)\, q_S(\theta_S(z))$.

Employed worker, no search ($s=0$). With no search, the only events are demographic exit ν, exogenous separation $\xi_S(z)$, and quality shocks $\lambda_S(z)$. The worker HJB is

$$\begin{aligned}(r+\nu+\xi_S(z)+\lambda_S(z))\,E^0_S(x,p,z) &= w^0_S(x,p,z) + \xi_S(z)\,U_S(x,z)\\ &\quad + \lambda_S(z)\left[U_S(x,z)\,\Gamma_z(p^*_S(x,z)) + \int_{p^*_S(x,z)}^{1} E_S(x,p',z)\,\mathrm{d}\Gamma_z(p')\right].\end{aligned} \tag{18}$$

Employed worker, search ($s=1$). Searching costs σ_S per unit time and generates outside contacts at rate $\kappa_S(z)$. Upon an offer $\tilde p$, the worker quits iff $\tilde p > p$ and the offered match is feasible. Hence

$$\begin{aligned}(r+\nu+\xi_S(z)+\lambda_S(z))\,E^1_S(x,p,z) &= (w^1_S(x,p,z) - \sigma_S) + \xi_S(z)\,U_S(x,z)\\ &\quad + \lambda_S(z)\left[U_S(x,z)\,\Gamma_z(p^*_S(x,z)) + \int_{p^*_S(x,z)}^{1} E_S(x,p',z)\,\mathrm{d}\Gamma_z(p')\right]\\ &\quad + \kappa_S(z)\int_{\max\{p,p^*_S(x,z)\}}^{1} \big(E_S(x,\tilde p,z) - E^1_S(x,p,z)\big)\,\mathrm{d}\Gamma_z(\tilde p).\end{aligned} \tag{19}$$

Firm values. A filled job yields flow profit $P_S(z)\,x\,p - w_S^s(x,p,z)$. With no search ($s=0$), there is no quit hazard. With OJS ($s=1$), the job is lost at rate $\kappa_S(z)(1-\Gamma_z(p))$ because any offer above p triggers a quit. Thus:

$$\begin{aligned}(r+\nu+\xi_S(z)+\lambda_S(z))\,J_S^0(x,p,z) &= P_S(z)\,x\,p - w_S^0(x,p,z)\\ &\quad + \lambda_S(z)\int_{p_S^*(x,z)}^1 J_S(x,p',z)\,\mathrm{d}\Gamma_z(p'),\end{aligned} \tag{20}$$

$$\begin{aligned}(r+\nu+\xi_S(z)+\lambda_S(z)+\kappa_S(z)(1-\Gamma_z(p)))\,J_S^1(x,p,z) &= P_S(z)\,x\,p - w_S^1(x,p,z)\\ &\quad + \lambda_S(z)\int_{p_S^*(x,z)}^1 J_S(x,p',z)\,\mathrm{d}\Gamma_z(p').\end{aligned} \tag{21}$$

8.4 Policy rules: endogenous separation and OJS cutoff

Endogenous separation cutoff. Define overall surplus as

$$S_S(x,p,z) \equiv \max\{S_S^0(x,p,z),\ S_S^1(x,p,z)\}$$

. Endogenous separation is characterised by a cutoff $p_S^*(x,z)$ satisfying

$$S_S(x,p_S^*(x,z),z) = 0, \qquad S_S(x,p,z) \geq 0 \iff p \geq p_S^*(x,z). \tag{22}$$

OJS cutoff (search vs no-search). The worker searches iff it is optimal:

$$s^*(x,p,z) = \mathbf{1}\{E_S^1(x,p,z) \geq E_S^0(x,p,z)\}.$$

Under monotonicity in p, there exists an OJS cutoff $p_S^{\mathrm{oj}}(x,z) \geq p_S^*(x,z)$ such that

$$E_S^1(x,p,z) \geq E_S^0(x,p,z) \iff p < p_S^{\mathrm{oj}}(x,z), \qquad E_S^1(x,p_S^{\mathrm{oj}}(x,z),z) = E_S^0(x,p_S^{\mathrm{oj}}(x,z),z). \tag{23}$$

Workers in low-quality matches search (the ladder gain exceeds σ_S); workers in high-quality matches do not.

8.5 Surplus and Wages

Step 1: useful identity. Because $E_S(x,p,z) = U_S(x,z) + \beta_S S_S(x,p,z)$ and $J_S(x,p,z) = (1-\beta_S)S_S(x,p,z)$ under the optimal regime at (x,p,z), we have

$$E_S(x,p,z) + J_S(x,p,z) = U_S(x,z) + S_S(x,p,z).$$

Hence, for any lower bound $\underline{p} \geq p_S^*(x,z)$,

$$\int_{\underline{p}}^1 (E_S(x,p',z) + J_S(x,p',z))\,\mathrm{d}\Gamma_z(p') = (1-\Gamma_z(\underline{p}))\,U_S(x,z) + \int_{\underline{p}}^1 S_S(x,p',z)\,\mathrm{d}\Gamma_z(p').$$

Step 2: eliminate wages. Adding the worker and firm HJBs in each search regime s and subtracting the unemployment HJB eliminates wages, leaving an HJB for S_S^s.

Surplus with no search ($s=0$).

$$\begin{aligned}(r+\nu+\xi_S(z)+\lambda_S(z))\,S_S^0(x,p,z) &= P_S(z)\,x\,p - b_S - (r+\nu)\,U_S(x,z)\\ &\quad + (\lambda_S(z) - \beta_S\kappa_S(z))\int_{p_S^*(x,z)}^1 S_S(x,p',z)\,\mathrm{d}\Gamma_z(p').\end{aligned} \tag{24}$$

Surplus with on-the-job search ($s=1$).

$$
\begin{aligned}
&(r+\nu+\xi_S(z)+\lambda_S(z)+\kappa_S(z)(1-\Gamma_z(p)))\,S_S^1(x,p,z)=P_S(z)\,x\,p-b_S-(r+\nu)\,U_S(x,z)-\sigma_S\\
&\quad+\lambda_S(z)\int_{p_S^*(x,z)}^1 S_S(x,p',z)\,\mathrm{d}\Gamma_z(p')-\beta_S\kappa_S(z)\int_{p_S^*(x,z)}^p S_S(x,p',z)\,\mathrm{d}\Gamma_z(p'). \qquad (25)
\end{aligned}
$$

8.6 Wage schedules

No OJS ($s=0$). Multiplying (24) by $(1-\beta_S)$ and comparing with the firm HJB (20):

$$
w_S^0(x,p,z)=\beta_S\,P_S(z)\,x\,p+(1-\beta_S)\,b_S+\beta_S(1-\beta_S)\,\kappa_S(z)\underbrace{\int_{p_S^*(x,z)}^1 S_S(x,p',z)\,\mathrm{d}\Gamma_z(p')}_{=:\,I(x,z)}. \qquad (26)
$$

The wage is linear in p for fixed x: the third term is an x-specific constant that reflects the unemployment-contact option available to the worker (weight β_S) and surrendered to the firm (weight $1-\beta_S$) when the match is valued under no search.

OJS ($s=1$). The same procedure applied to (25) and (21):

$$
w_S^1(x,p,z)=\beta_S\,P_S(z)\,x\,p+(1-\beta_S)(b_S+\sigma_S)+\beta_S(1-\beta_S)\,\kappa_S(z)\underbrace{\int_{p_S^*(x,z)}^p S_S(x,p',z)\,\mathrm{d}\Gamma_z(p')}_{=:\,I_{\text{low}}(x,p,z)}. \qquad (27)
$$

OJS wage premium. Comparing (27) and (26), the OJS wage premium is

$$
w_S^1(x,p,z)-w_S^0(x,p,z)=-(1-\beta_S)\,\sigma_S-\beta_S(1-\beta_S)\,\kappa_S(z)\int_p^1 S_S(x,p',z)\,\mathrm{d}\Gamma_z(p'). \qquad (28)
$$

The premium is *non-positive*: the search cost $(1-\beta_S)\sigma_S$ is borne entirely by the worker (Nash shifts it from w_S^1 relative to w_S^0), while the ladder option term $-\beta_S(1-\beta_S)\kappa_S\int_p^1 S_S\,\mathrm{d}\Gamma_z\leq 0$ reflects the firm's compensation by a wage discount proportional to the expected capital gain it would lose to a poaching offer. Both terms vanish as $p\to 1$ (no further upward ladder) and are largest near $p_S^*(x,z)$.

8.7 Vacancies, poaching, and free entry

Posting a skilled vacancy costs $k_S(z)>0$ per unit time. A vacancy meets a *random job seeker* at rate $q_S(\theta_S(z))$. Since job seekers include unemployed and employed searchers, the expected value of a contact depends on the composition of seekers.

Define the density of job seekers (unemployed + employed searchers) as

$$
\widetilde{u}_S(x,z)\equiv u_S(x,z)+\int_0^1 s^*(x,p,z)\,e_S(x,p,z)\,\mathrm{d}p, \qquad \widetilde{\mathcal{U}}_S(z)=\int_0^1 \widetilde{u}_S(x,z)\,\mathrm{d}x.
$$

Conditional on a contact, the worker type is drawn from $\widetilde{u}_S(\cdot,z)$, and:

- if the contacted seeker is unemployed, the match quality draw is $\tilde{p}\sim\Gamma_z$ and the match is formed iff $\tilde{p}\geq p_S^*(x,z)$, yielding value $J_S(x,\tilde{p},z)$;
- if the contacted seeker is an employed searcher currently at quality p, the offer is $\tilde{p}\sim\Gamma_z$ and the worker accepts (quits) iff $\tilde{p}>p$ and $\tilde{p}\geq p_S^*(x,z)$, yielding value $J_S(x,\tilde{p},z)$.

Thus the vacancy HJB is

$$
rV_S(z)=-k_S(z)+q_S(\theta_S(z))(\mathcal{J}_S(z)-V_S(z)), \qquad (29)
$$

where the expected surplus from a random contact is

$$\mathcal{J}_S(z) \equiv \frac{1}{\widetilde{\mathcal{U}}_S(z)} \underbrace{\int_0^1 u_S(x,z) \int_{p_S^*(x,z)}^1 J_S(x,\tilde{p},z)\,\mathrm{d}\Gamma_z(\tilde{p})\,\mathrm{d}x}_{\text{meeting unemployed}}$$
$$+ \underbrace{\int_0^1 \int_0^1 s^*(x,p,z)\,e_S(x,p,z) \int_{\max\{p,p_S^*(x,z)\}}^1 J_S(x,\tilde{p},z)\,\mathrm{d}\Gamma_z(\tilde{p})\,\mathrm{d}p\,\mathrm{d}x}_{\text{poaching employed searchers}}\,. \tag{30}$$

Free entry implies $V_S(z) = 0$, hence

$$q_S(\theta_S(z))\,\mathcal{J}_S(z) = k_S(z). \tag{31}$$

Given (u_S, e_S, s^*), this pins down $\theta_S(z)$ by inverting q_S.

8.8 Stationary cross-sectional distributions (trained workers)

Let $u_S(x,z)$ be the density of skilled unemployed and $e_S(x,p,z)$ the density of employed trained workers at quality p. Accounting within the trained population:

$$m_S(x,z) = u_S(x,z) + \int_{p_S^*(x,z)}^1 e_S(x,p,z)\,\mathrm{d}p.$$

Endogenous destruction from quality shocks. Quality shocks arrive at rate $\lambda_S(z)$ and redraw $p' \sim \Gamma_z$. A match is destroyed if $p' < p_S^*(x,z)$, so the endogenous destruction hazard is

$$\delta_S^{\text{end}}(x,z) \equiv \lambda_S(z)\,\Gamma_z(p_S^*(x,z)).$$

Job-to-job quit hazard under OJS. If the worker searches ($s^*(x,p,z) = 1$), outside offers arrive at rate $\kappa_S(z)$ and the worker quits iff $\tilde{p} > p$, with probability $1 - \Gamma_z(p)$. Hence

$$\delta_S^{\text{jj}}(x,p,z) \equiv s^*(x,p,z)\,\kappa_S(z)\,(1 - \Gamma_z(p)).$$

Unemployment balance. For each x:

$$\Big(\nu + \kappa_S(z)(1 - \Gamma_z(p_S^*(x,z)))\Big)\,u_S(x,z) = \phi\,t(x,z) + \int_{p_S^*(x,z)}^1 (\xi_S(z) + \delta_S^{\text{end}}(x,z))\,e_S(x,p,z)\,\mathrm{d}p. \tag{32}$$

Employment-by-quality balance. For each x and each $p \geq p_S^*(x,z)$, the density $e_S(x,p,z)$ satisfies a steady-state balance: outflows from the state (x,p) equal inflows into (x,p). Outflows are due to demographic exit, exogenous separation, quality shocks, and job-to-job quits. Inflows come from three sources: (i) hiring from unemployment into quality p (offer density $\gamma_z(p)$), (ii) quality shocks from any employed state into p (redraw density $\gamma_z(p)$), and (iii) job-to-job upgrades from worse jobs $p' < p$ when those workers search and draw p.

$$\Big(\nu + \xi_S(z) + \lambda_S(z) + \delta_S^{\text{jj}}(x,p,z)\Big)\,e_S(x,p,z)$$
$$= \underbrace{\kappa_S(z)\,u_S(x,z)\,\gamma_z(p)}_{\text{hiring from unemployment into } p} + \underbrace{\lambda_S(z)\,\gamma_z(p) \int_{p_S^*(x,z)}^1 e_S(x,p',z)\,\mathrm{d}p'}_{\text{quality shocks into } p}$$
$$+ \underbrace{\kappa_S(z)\,\gamma_z(p) \int_{p_S^*(x,z)}^p s^*(x,p',z)\,e_S(x,p',z)\,\mathrm{d}p'}_{\text{job-to-job inflow from worse jobs}}\,. \tag{33}$$

Remark 2 (Implementation). *On a grid $p_1 < \cdots < p_{N_p}$, equation (33) becomes a linear system in the unknown vector $\{e_S(x, p_i, z)\}_{i=1}^{N_p}$ for each x: the quit hazard contributes to the diagonal; quality shocks add a rank-one inflow term proportional to $\gamma_z(p_i)$; and the job-ladder inflow term is lower-triangular because it depends only on $\{e_S(x, p_j, z)\}_{j<i}$ in the OJS region ($s^* = 1$).*

9 Skilled-block equilibrium and nested fixed-point algorithm

Fix a regime $z \in \mathcal{Z}$ and take the trained population density $m_S(\cdot, z)$ as given from the unskilled block. A *skilled-block equilibrium* is a collection

$$\{\theta_S(z),\ U_S(\cdot,z),\ E_S^0(\cdot,\cdot,z),\ E_S^1(\cdot,\cdot,z),\ J_S^0(\cdot,\cdot,z),\ J_S^1(\cdot,\cdot,z),\ w_S^0(\cdot,\cdot,z),\ w_S^1(\cdot,\cdot,z),$$

$$p_S^*(\cdot,z),\ p_S^{\text{oj}}(\cdot,z),\ s^*(\cdot,\cdot,z),\ u_S(\cdot,z),\ e_S(\cdot,\cdot,z)\}$$

such that:

(S1) **Worker values.** Unemployment satisfies (17). Employed values satisfy (18)–(19), with $E_S(x,p,z) \equiv \max\{E_S^0(x,p,z),\ E_S^1(x,p,z)\}$.

(S2) **Firm values.** Filled-job values satisfy (20)–(21), with $J_S(x,p,z) \equiv J_S^{s^*(x,p,z)}(x,p,z)$.

(S3) **OJS policy.** The worker's search decision is optimal: $s^*(x,p,z) = \mathbf{1}\{E_S^1(x,p,z) \geq E_S^0(x,p,z)\}$, summarised by a cutoff $p_S^{\text{oj}}(x,z)$ satisfying (23).

(S4) **Endogenous separation.** The reservation cutoff $p_S^*(x,z)$ satisfies (22).

(S5) **Wage determination.** Wages satisfy Nash bargaining (16). (Equivalently, w_S^s are recovered from the worker HJBs (18)–(19) and checked/adjusted to enforce (16) pointwise.)

(S6) **Tightness and free entry (Chapter 4 congestion).** Tightness is defined using job seekers (unemployed plus employed searchers) and free entry holds: $q_S(\theta_S(z))\,\mathcal{J}_S(z) = k_S(z)$.

(S7) **Stationary distributions.** Given $(\theta_S, p_S^*, p_S^{\text{oj}}, s^*)$, the stationary densities (u_S, e_S) satisfy (32)–(33) and the accounting identity $m_S(x,z) = u_S(x,z) + \int_{p_S^*(x,z)}^1 e_S(x,p,z)\,\mathrm{d}p$.

9.1 Nested fixed-point algorithm (skilled side; fixed regime z)

For computation, it is convenient to organise the skilled block as a nested fixed point. The *inner loop* solves the value functions (and wage schedules) given outer objects. The *outer loop* updates $(\theta_S, p_S^*, p_S^{\text{oj}})$ and the stationary composition until all equilibrium conditions hold jointly.

Outer-loop unknowns. Iterate on

$$\Big(\theta_S^{(n)}(z),\ p_S^{*,(n)}(\cdot,z),\ p_S^{\text{oj},(n)}(\cdot,z)\Big).$$

Step 1. Initialise (outer loop). Choose initial guesses $\theta_S^{(0)}(z) > 0$ and feasible cutoffs $p_S^{*,(0)}(x,z) \in (0,1)$ and $p_S^{\text{oj},(0)}(x,z) \in [p_S^{*,(0)}(x,z), 1]$. Initialise stationary densities $(u_S^{(0)}, e_S^{(0)})$ consistent with $m_S(\cdot,z)$ (e.g. set $u_S^{(0)} = \kappa\, m_S$ and allocate the remainder of mass to the top p grid point).

Step 2. Inner loop: surplus iteration (given outer objects). Holding $(\theta_S^{(n)}, p_S^{*,(n)}, p_S^{\text{oj},(n)})$ fixed, iterate on (U_S, S_S^0, S_S^1) until convergence. Let $\kappa_S = \theta_S^{(n)} q_S(\theta_S^{(n)})$ throughout. One inner cycle for each x:

- **Step 2.1. Tail integral.** Let j_0 be the first grid index with $p_{j_0} \geq p_S^{*,(n)}(x,z)$. Compute the tail integral using the *previous* iteration's surplus: $I^{(m)}(x,z) = \int_{p_S^{*,(n)}(x,z)}^1 S_S^{\max,(m)}(x,p',z)\,\mathrm{d}\Gamma_z(p')$, where $S_S^{\max,(m)} = \max\{S_S^{0,(m)}, S_S^{1,(m)}\}$.

This is approximated by accumulating the weighted sum $\sum_{k\geq j_0} S_S^{\max,(m)}(x,p_k,z)\,w_\Gamma(p_k)$ in a backward pass over the p-grid.

- **Step 2.2. Unemployment value (closed form).**

$$U_S^{(m+1)}(x,z) = \frac{b_S + \kappa_S\,\beta_S\,I^{(m)}(x,z)}{r+\nu}.$$

- **Step 2.3. Surplus on p-grid (closed-form evaluation).** For each grid point p_j:
  - If $j < j_0$ (infeasible region): $S_S^{0,(m+1)} = S_S^{1,(m+1)} = 0$.
  - If $j \geq j_0$:

$$S_S^{0,(m+1)}(x,p_j,z) = \frac{P_S(z)\,x\,p_j - (r+\nu)\,U_S^{(m+1)} + \lambda_S(z)\,I^{(m)}}{r+\nu+\xi_S(z)+\lambda_S(z)},$$

$$S_S^{1,(m+1)}(x,p_j,z) = \frac{P_S(z)\,x\,p_j - (r+\nu)\,U_S^{(m+1)} - \sigma_S + \lambda_S(z)\,I^{(m)} + \kappa_S\,\beta_S\int_{p_j}^1 S_S^{\max,(m)}\,\mathrm{d}\Gamma_z}{r+\nu+\xi_S(z)+\lambda_S(z)+\kappa_S(1-\Gamma_z(p_j))}.$$

  The numerator integral $\int_{p_j}^1 S_S^{\max,(m)}\,\mathrm{d}\Gamma_z$ is available from Step 2.1.

- **Step 2.4. Nash split.** Recover employed values and firm values via $E_S^{s,(m+1)} = U_S^{(m+1)} + \beta_S S_S^{s,(m+1)}$, $J_S^{s,(m+1)} = (1-\beta_S)S_S^{s,(m+1)}$, $s\in\{0,1\}$. Wages are not computed explicitly.

Stop when $\sup_{x,p}\max\{|S_S^{0,(m+1)} - S_S^{0,(m)}|,\ |S_S^{1,(m+1)} - S_S^{1,(m)}|,\ |U_S^{(m+1)} - U_S^{(m)}|\} <$ tol_inner.

Step 3. Update cutoffs (outer update).

- **Step 3.1. Separation cutoff p_S^*.** Scan forward from the previous cutoff index j_0^{prev} for the first grid point where $S_S^{\max,(m+1)}(x,p_j,z)\geq 0$: $p_S^{*,\text{prop}}(x,z) = p_{j^*}$. Apply damped update with damping coefficient $\omega\in(0,1)$: $p_S^{*,(n+1)}(x,z) = \omega\,p_S^{*,\text{prop}}(x,z) + (1-\omega)\,p_S^{*,(n)}(x,z)$.
- **Step 3.2. OJS cutoff p_S^{oj}.** Scan forward from j_0 for the first grid point where $S_S^{1,(m+1)}(x,p_j,z) - S_S^{0,(m+1)}(x,p_j,z)\leq 0$: $p_S^{\text{oj},(n+1)}(x,z) = p_{j^{\text{oj}}}$. Apply the same damped update: $p_S^{\text{oj},(n+1)}(x,z) = \omega\,p_S^{\text{oj,prop}} + (1-\omega)\,p_S^{\text{oj},(n)}(x,z)$. The OJS policy follows as $s^*(x,p,z) = \mathbf{1}\{p < p_S^{\text{oj},(n+1)}(x,z)\}$.

Step 4. Update stationary distributions (composition block). Given $(\theta_S^{(n)}, p_S^{*,(n+1)}, p_S^{\text{oj},(n+1)})$, solve the stationary flow-balance system (32)–(33) (on the discretised p grid) subject to the accounting identity for each x. From the solution, compute the seeker density

$$\tilde{u}_S(x,z) = u_S(x,z) + \int_0^1 s^*(x,p,z)\,e_S(x,p,z)\,\mathrm{d}p, \qquad \tilde{\mathcal{U}}_S(z) = \int_0^1 \tilde{u}_S(x,z)\,\mathrm{d}x.$$

Step 5. Update tightness by free entry (outer update). Compute $\mathcal{J}_S(z)$ from (30) using the updated (u_S, e_S, s^*) and the converged $J_S(x,p,z)$. Update θ_S using free entry (31):

$$q_S(\theta_S^{(n+1)}(z)) = \frac{k_S(z)}{\mathcal{J}_S(z)}.$$

Invert $q_S(\theta) = \mu_S\,\theta^{-\eta_S}$ to obtain $\theta_S^{(n+1)}(z)$. Use Anderson acceleration when updating tightness, to reduce fluctuations.

Step 6. Convergence (outer loop). Stop if

$$\max\Big\{|\theta_S^{(n+1)}(z) - \theta_S^{(n)}(z)|,\ \|p_S^{*,(n+1)}(\cdot,z) - p_S^{*,(n)}(\cdot,z)\|_\infty,\ \|p_S^{\text{oj},(n+1)}(\cdot,z) - p_S^{\text{oj},(n)}(\cdot,z)\|_\infty\Big\} < \text{tol}.$$

Otherwise set $n \leftarrow n+1$ and return to Step 2.

Link to the unskilled block. On convergence, the skilled unemployment value $U_S(x,z)$ feeds back into the training value equation (1), while the trained density $m_S(\cdot,z)$ is taken as given from training completion in the unskilled block. A global equilibrium is obtained by iterating between the two blocks until these link variables are jointly consistent.

10 Full model equilibrium and global algorithm

Fix a regime $z \in \mathcal{Z}$. A *full equilibrium* in regime z consists of unskilled-block and skilled-block objects such that:

(F1) The unskilled block satisfies conditions (E1)–(E5) given the skilled unemployment value $U_S(\cdot,z)$.

(F2) The skilled block satisfies conditions (S1)–(S7) given the trained population density $m_S(x,z) = \frac{\phi}{\nu}\, t(x,z)$.

(F3) The link variables are consistent: $U_S(\cdot,z)$ used in the unskilled block equals the skilled-block solution, and $m_S(\cdot,z)$ used in the skilled block equals training completion from the unskilled block.

10.1 Global fixed-point algorithm (fixed regime z)

The full model is solved via an outer fixed point over the link variables $(U_S(\cdot,z), m_S(\cdot,z))$:

Step 1. Initialise. Choose initial guesses $U_S^{(0)}(\cdot,z)$ and $m_S^{(0)}(\cdot,z)$.

Step 2. Solve the unskilled block. Given $U_S^{(n)}(\cdot,z)$, solve the unskilled equilibrium as in Section 7. Obtain $(t^{(n)}(\cdot,z), u_U^{(n)}(\cdot,z), \theta_U^{(n)}(z), \ldots)$. Update the implied trained mass: $m_S^{\text{new}}(x,z) = \frac{\phi}{\nu}\, t^{(n)}(x,z)$.

Step 3. Solve the skilled block. Given $m_S^{\text{new}}(\cdot,z)$, solve the skilled equilibrium using the nested algorithm described in Section 9. Obtain $U_S^{\text{new}}(\cdot,z)$ and $\theta_S^{(n)}(z)$.

Step 4. Update link variables. Update U_S and m_S.

Step 5. Convergence. Stop if

$$\max\{\|U_S^{(n+1)} - U_S^{(n)}\|_\infty,\ \|m_S^{(n+1)} - m_S^{(n)}\|_\infty\} < \text{tol}.$$

Otherwise set $n \leftarrow n+1$ and return to Step 2.

11 Transition dynamics: evolution of densities after a regime change

Consider an unexpected, permanent regime switch at calendar time T from z_0 to z_1. At $t = T^+$, cross-sectional densities are predetermined by the pre-switch history. For $t > T$, all hazards and policies are those of regime z_1.

Equilibrium object. A transition equilibrium is a collection of time paths for value functions, policies, market tightness, and cross-sectional distributions such that:

1. Value functions solve time-dependent HJB equations given $(\theta_U(t), \theta_S(t))$;
2. Policies $(\tau, p_U^*, p_S^*, p_S^{\text{oj}})$ are optimal pointwise;
3. Distributions solve the laws of motion forward in time;

4. Market tightness satisfies free entry at each date;
5. Initial distributions match the z_0 steady state and the economy converges to the z_1 steady state.

Throughout this section, fix $z = z_1$ and suppress z in notation when convenient.

11.1 Time-dependent objects and hazards

All objects are time-dependent for $t > T$ and are evaluated under the post-switch regime z_1.

Let:
- $u_U(x,t)$: untrained unemployed,
- $t(x,t)$: workers in training,
- $u_S(x,t)$: skilled unemployed,
- $e_S(x,p,t)$: skilled employment by quality.

Market tightness is time-dependent:

$$\theta_U(t), \qquad \theta_S(t),$$

with job-finding rates

$$f_U(t) = \theta_U(t) q_U(\theta_U(t)), \qquad f_S(t) = \theta_S(t) q_S(\theta_S(t)).$$

Policies are also time-dependent:

$$\tau(x,t), \quad p_U^*(x,t), \quad p_S^*(x,t), \quad p_S^{\mathrm{oj}}(x,t).$$

Endogenous destruction hazards.

$$\delta_U(x,t) = \lambda_U(z_1)\, G_{z_1}(p_U^*(x,t)), \qquad \delta_S^{\mathrm{end}}(x,t) = \lambda_S(z_1)\, \Gamma_{z_1}(p_S^*(x,t)).$$

11.2 Segment accounting identities

Define the untrained (unskilled-segment) density by type:

$$m_U(x,t) \equiv u_U(x,t) + t(x,t) + e_U(x,t),$$

where $e_U(x,t)$ denotes untrained unskilled employment density (not tracked as a separate state variable). Thus, $e_U(x,t) = m_U(x,t) - u_U(x,t) - t(x,t)$.

Define the trained (skilled-segment) density by type:

$$m_S(x,t) \equiv u_S(x,t) + e_S^{\mathrm{tot}}(x,t),$$

where $e_S^{\mathrm{tot}}(x,t) \equiv \int_{p_S^*(x)}^1 e_S(x,p,t)\, \mathrm{d}p$ is total skilled employment mass of type x.

Aggregate segment masses are

$$M_U(t) \equiv \int_0^1 m_U(x,t)\, \mathrm{d}x, \qquad M_S(t) \equiv \int_0^1 m_S(x,t)\, \mathrm{d}x.$$

11.3 Laws of motion: unskilled segment

Training stock. Training receives inflow from untrained unemployment at rate $\tau(x)$ and loses mass through completion at rate ϕ and demographic exit at rate ν:

$$\partial_t\, t(x,t) = \tau(x)\, u_U(x,t) - (\phi + \nu)\, t(x,t). \tag{34}$$

Untrained unemployment. Untrained unemployment receives inflow from demographic replacement and from endogenous job destruction out of unskilled employment. It loses mass due to job finding, training entry, and demographic exit:

$$\partial_t\, u_U(x,t) = \nu\, \ell(x) + \delta_U(x)\, e_U(x,t) - \big(f_U + \tau(x) + \nu\big)\, u_U(x,t), \tag{35}$$

where $e_U(x,t) = m_U(x,t) - u_U(x,t) - t(x,t)$.

Untrained-segment mass by type. Because training completion moves workers from the unskilled to the skilled segment, the untrained mass $m_U(x,t)$ evolves as

$$\partial_t\, m_U(x,t) = \nu\, \ell(x) - \nu\, m_U(x,t) - \phi\, t(x,t). \tag{36}$$

Training entry reshuffles mass within the unskilled segment, so it does not appear in $\partial_t m_U$; completion removes mass from m_U.

11.4 Laws of motion: skilled segment

Skilled unemployment. Skilled unemployment receives inflow from training completion and from separations out of skilled employment (exogenous separations and endogenous destruction). It loses mass due to job finding into acceptable matches and demographic exit:

$$\partial_t\, u_S(x,t) = \phi\, t(x,t) + (\xi_S(z) + \delta_S^{\text{end}}(x))\, e_S^{\text{tot}}(x,t) - \Big(\nu + f_S(1 - \Gamma_z(p_S^*(x)))\Big)\, u_S(x,t). \tag{37}$$

Trained-segment mass by type. The trained mass $m_S(x,t)$ increases via training completion and decreases via demographic exit:

$$\partial_t\, m_S(x,t) = \phi\, t(x,t) - \nu\, m_S(x,t). \tag{38}$$

(Job-to-job moves reshuffle employment across firms and match qualities but do not change segment membership, hence do not enter $\partial_t m_S$.)

11.5 Aggregate mass dynamics

Integrating (36)–(38) over $x \in [0,1]$ yields the aggregate segment mass dynamics:

$$\dot{M}_U(t) = \nu \int_0^1 \ell(x)\,\mathrm{d}x - \nu M_U(t) - \phi \int_0^1 t(x,t)\,\mathrm{d}x, \tag{39}$$

$$\dot{M}_S(t) = \phi \int_0^1 t(x,t)\,\mathrm{d}x - \nu M_S(t). \tag{40}$$

Since $\ell(x)$ integrates to 1, then $\dot{M}_U(t) = \nu - \nu M_U(t) - \phi\, \mathcal{T}(t)$, $\dot{M}_S(t) = \phi\, \mathcal{T}(t) - \nu M_S(t)$, where $\mathcal{T}(t) \equiv \int_0^1 t(x,t)\,\mathrm{d}x$.

Adding shows total population is constant: $\frac{d}{dt}(M_U(t) + M_S(t)) = \nu - \nu(M_U(t) + M_S(t))$, so if $M_U(T) + M_S(T) = 1$ then $M_U(t) + M_S(t) = 1$ for all $t \geq T$.

11.6 Initial condition at the switch

Let $(u_U^0(x), t^0(x), u_S^0(x), m_U^0(x), m_S^0(x))$ denote the pre-switch stationary densities under z_0. At the instant of the switch,

$$u_U(x,T^+) = u_U^0(x), \quad t(x,T^+) = t^0(x), \quad u_S(x,T^+) = u_S^0(x),$$

and segment masses satisfy

$$m_U(x,T^+) = m_U^0(x), \quad m_S(x,T^+) = m_S^0(x).$$

For $t > T$, the system (34)–(38) evolves forward under the post-switch objects $(\tau, p^*, p_S^*, \theta_U, \theta_S)$ implied by regime z_1.

11.7 Market tightness along the transition

At each date t, market tightness is determined by free entry given the current distributions and continuation values.

Unskilled market.

$$q_U(\theta_U(t)) = \frac{k_U(z_1)\,\mathcal{U}_U(t)}{\int_0^1 J_U(x,1,t)\,u_U(x,t)\,\mathrm{d}x},$$

where $\mathcal{U}_U(t) = \int_0^1 u_U(x,t)\,\mathrm{d}x$.

Skilled market.

$$q_S(\theta_S(t)) = \frac{k_S(z_1)}{\mathcal{J}_S(t)},$$

where $\mathcal{J}_S(t)$ is the expected value of a contact with a random job seeker under the current distribution.

Inversion.

$$\theta_j(t) = \left(\frac{\mu_j}{q_j(t)}\right)^{1/\eta_j}, \qquad j \in \{U,S\}.$$

11.8 Transition mapping

At each date t, the equilibrium mapping is:

$$(u_U, t, u_S, e_S)_t \implies (\theta_U, \theta_S)_t \implies \text{value functions} \implies (\tau, p_U^*, p_S^*, p_S^{\mathrm{oj}})_t \implies \partial_t(u_U, t, u_S, e_S).$$

11.9 Exact transition equilibrium: numerical algorithm

The transition path is obtained as a deterministic equilibrium in which forward-looking value functions and forward-moving distributions are jointly consistent. This is a forward–backward fixed point.

Time discretisation. Fix a horizon $T_{\max}$ large enough that the economy is arbitrarily close to the stationary equilibrium of regime z_1. Discretise time:

$$0 = t_0 < t_1 < \cdots < t_N = T_{\max}.$$

Step 0. Boundary objects. Compute the stationary equilibria under z_0 and z_1.

- Initial condition (at $t = 0$): distributions equal the z_0 steady state.
- Terminal condition (at $t = T_{\max}$): value functions equal the z_1 steady state.

Step 1. Initial guess. Choose initial paths for market tightness:

$$\{\theta_U^{(0)}(t_n),\ \theta_S^{(0)}(t_n)\}_{n=0}^N.$$

Step 2. Backward pass: value functions and policies. Given $(\theta_U^{(k)}, \theta_S^{(k)})$, solve the time-dependent HJB system backward from t_N to t_0.

At each date t_n:

- Solve for $(U_S, T, U_U^{\mathrm{search}}, U_U)$.
- Solve match values $(E_U, J_U, E_S^0, E_S^1, J_S^0, J_S^1)$.
- Update policies:

$$\tau(x,t_n), \quad p_U^*(x,t_n), \quad p_S^*(x,t_n), \quad p_S^{\mathrm{oj}}(x,t_n).$$

Step 3. Forward pass: distributions. Given policy paths, solve the laws of motion forward from t_0 to t_N:

$$(u_U, t, u_S, e_S)_{t_0} \longrightarrow (u_U, t, u_S, e_S)_{t_N}.$$

Step 4. Update tightness paths. At each date t_n, update tightness using free entry:

$$q_U(\theta_U^{\text{new}}(t_n)) = \frac{k_U(z_1)\,\mathcal{U}_U(t_n)}{\int_0^1 J_U(x,1,t_n)\,u_U(x,t_n)\,\mathrm{d}x},$$

$$q_S(\theta_S^{\text{new}}(t_n)) = \frac{k_S(z_1)}{\mathcal{J}_S(t_n)}.$$

Invert $q_j(\theta) = \mu_j \theta^{-\eta_j}$ to obtain $\theta_j^{\text{new}}(t_n)$.

Step 5. Convergence. Iterate Steps 2–4 until

$$\max_n \Big(|\theta_U^{(k+1)}(t_n) - \theta_U^{(k)}(t_n)|,\ |\theta_S^{(k+1)}(t_n) - \theta_S^{(k)}(t_n)| \Big) < \text{tol}.$$

The resulting paths constitute the transition equilibrium.

12 Model Properties

This section collects key qualitative properties of the model that shape the economic interpretation of the estimation results and the transition-dynamics exercises.

12.1 Training cutoff and the education margin

Proposition 1 (Training cutoff)**.** *Fix a regime $z \in \mathcal{Z}$. Under the maintained functional forms ($c'(x) < 0$, U_S and T continuous in x), the net training gain*

$$\Delta_T(x,z) \;=\; -\,c(x) + T(x,z) - U_U^{\text{search}}(x,z)$$

is increasing in x. Hence the optimal training policy is a cutoff rule:

$$\tau(x,z) = \mathbf{1}\{x > \bar{x}(z)\}, \qquad \Delta_T(\bar{x}(z), z) = 0.$$

Workers with innate type $x > \bar{x}(z)$ train upon entering unskilled unemployment; those with $x \le \bar{x}(z)$ search in the unskilled market.

Intuition. Higher-ability workers benefit from training through two channels: (i) the direct cost $c(x) = e^c\,(1-x)\,e^{-x}$ is decreasing in x, and (ii) the return to skilled-market access, captured by $T(x,z)$, is increasing in x because skilled output $P_S(z)\,x\,p$ rises with ability.

Comparative statics. The cutoff $\bar{x}(z)$ responds to regime changes through several channels:

1. A rise in $P_S(z)$ raises $T(x,z)$ for all x, lowering $\bar{x}$ (more workers train).
2. A rise in $P_U(z)$ raises U_U^{search} through higher unskilled wages, raising $\bar{x}$ (fewer workers train).
3. A rise in unskilled tightness $\theta_U(z)$ raises the unskilled job-finding rate and hence U_U^{search}, also pushing $\bar{x}$ up.
4. A rise in skilled tightness $\theta_S(z)$ raises the expected return to training $T(x,z)$, pushing $\bar{x}$ down.

In a general-equilibrium crisis, all four channels operate simultaneously. The net effect on $\bar{x}$ depends on *which* parameters shift and by how much—precisely what the multi-regime estimation recovers.

12.2 Separation thresholds

Proposition 2 (Unskilled separation cutoff)**.** *For each type x and regime z, there exists a unique reservation match quality $p^*(x,z) \in (0,1)$ satisfying $J_U(x,p^*,z)=0$, with the closed-form expression*

$$p^*(x,z) = \frac{(r+\nu)\,U_U(x,z) - \lambda_U(z)\,\int_{p^*}^1 S_U(x,p',z)\,\mathrm{d}G_z(p')}{P_U(z)\,x},$$

The cutoff is decreasing in x: higher-ability workers tolerate lower match quality because the surplus $P_U(z)\,x\,p$ grows faster in p for large x.

Proposition 3 (Skilled separation and OJS cutoffs)**.** *For each type x and regime z:*

1. *There exists a separation cutoff $p_S^*(x,z)$ satisfying $S_S(x,p_S^*,z)=0$. Matches with $p<p_S^*$ are destroyed upon a quality shock.*
2. *There exists an OJS cutoff $p_S^{\mathrm{oj}}(x,z) \geq p_S^*(x,z)$ satisfying $E_S^1(x,p_S^{\mathrm{oj}},z) = E_S^0(x,p_S^{\mathrm{oj}},z)$. Workers in matches with $p<p_S^{\mathrm{oj}}$ search on the job; those with $p \geq p_S^{\mathrm{oj}}$ do not.*

Both cutoffs are decreasing in x.

Intuition for the OJS cutoff. Search on the job is optimal when the expected capital gain from climbing the job ladder exceeds the flow search cost σ_S. Workers in low-quality matches have more to gain from a new draw $\tilde{p} \sim \Gamma_z$, so they search; those already in high-quality matches do not.

12.3 Wage structure

The model generates three wage schedules, all linear in match quality p for fixed type x:

$$w_U(x,p,z) = \beta_U\,P_U(z)\,x\,p + (1-\beta_U)(r+\nu)\,U_U(x,z), \tag{41}$$

$$w_S^0(x,p,z) = \beta_S\,P_S(z)\,x\,p + (1-\beta_S)\,b_S + \beta_S(1-\beta_S)\,\kappa_S(z)\,I(x,z), \tag{42}$$

$$w_S^1(x,p,z) = \beta_S\,P_S(z)\,x\,p + (1-\beta_S)(b_S+\sigma_S) + \beta_S(1-\beta_S)\,\kappa_S(z)\,I_{\mathrm{low}}(x,p,z). \tag{43}$$

Key properties of the wage structure:

1. **Within-group dispersion.** Wage dispersion within each skill group arises from heterogeneity in both x and p. The distributions of x (governed by a_ℓ, b_ℓ) and p (governed by α_U or a_Γ, b_Γ) jointly determine variance and skewness, which the SMM targets via higher-order wage moments.
2. **Skill premium.** The log skill premium $\mathbb{E}[\log w_S] - \mathbb{E}[\log w_U]$ depends on P_S/P_U, the match-quality distributions, bargaining weights, and the composition of employed workers. It is directly targeted by the SMM objective.
3. **OJS wage premium.** Comparing (43) and (42):

$$w_S^1 - w_S^0 = -(1-\beta_S)\,\sigma_S - \beta_S(1-\beta_S)\,\kappa_S \int_p^1 S_S(x,p',z)\,\mathrm{d}\Gamma_z(p') \;\leq\; 0.$$

Searching workers earn *less* than non-searchers at the same (x,p): Nash bargaining shifts the search cost and the expected poaching loss onto the worker.

12.4 Market tightness and free entry

Market tightness in each sector is pinned down by the free-entry condition. For the unskilled market:

$$q_U(\theta_U(z)) = \frac{k_U(z)\,\mathcal{U}_U(z)}{\displaystyle\int_0^1 J_U(x,1,z)\,u_U(x,z)\,\mathrm{d}x},$$

and for the skilled market:

$$q_S(\theta_S(z))\,\mathcal{J}_S(z) = k_S(z),$$

where $\mathcal{J}_S(z)$ is the expected surplus from a random contact (equation (30)), which depends on the composition of job seekers (unemployed plus on-the-job searchers).

Tightness responds to regime changes through two channels: (i) direct shifts in vacancy costs $k_j(z)$ and productivity $P_j(z)$ that change the surplus from a filled job, and (ii) composition effects through changes in the unemployed pool $u_j(\cdot,z)$. During the *transition* after a regime switch, tightness adjusts gradually because the composition of job seekers inherits the pre-switch distribution and evolves slowly.

12.5 Transition dynamics: adjustment speed

After an unexpected permanent regime switch from z_0 to z_1 at time T, the economy adjusts toward the z_1 steady state. The speed of adjustment is governed by three rates:

1. **Demographic turnover ν.** The fundamental "clock" of the model. New entrants arrive with density $\ell(x)$ and make training decisions under the new regime. At rate ν, the pre-switch cohort is gradually replaced.

2. **Training completion ϕ.** Workers in training complete at rate ϕ and enter the skilled segment. The pipeline of trainees adjusts on a timescale of roughly $1/(\phi+\nu)$.

3. **Job creation and destruction.** The job-finding rates f_U, f_S and separation hazards δ_U, δ_S govern how quickly the employed/unemployed composition adjusts within each segment. These are fast relative to training and demography.

The aggregate skilled share $M_S(t)$ satisfies

$$\dot{M}_S(t) = \phi\,\mathcal{T}(t) - \nu\,M_S(t),$$

where $\mathcal{T}(t) = \int_0^1 t(x,t)\,\mathrm{d}x$ is the total training stock. This is a stable linear ODE with convergence rate ν, implying a half-life of $\ln 2/\nu \approx 14$ months when $\nu = 0.05$ (monthly). Skilled-share adjustment is therefore slow and persistent—a key feature distinguishing this model from standard search models without a training margin.

Demand shock vs. supply/reallocation shock. The nature of the crisis matters for the transition path:

- A **demand shock** (e.g. the financial crisis) primarily reduces P_U and P_S, increasing separation and reducing vacancy creation. If P_U falls proportionally more than P_S, the relative return to training rises and $\bar{x}$ falls—more workers train.

- A **supply/reallocation shock** (e.g. COVID-19) may shift match-quality distributions (α_U, a_Γ, b_Γ), vacancy costs, and separation rates in ways that differ qualitatively from a demand shock. Whether training incentives rise or fall depends on which parameters shift and how the two markets are differentially affected.

The multi-regime estimation identifies exactly which parameters change, and the transition-dynamics computation traces out the implied adjustment path—allowing a structural decomposition of the crisis into its component channels.